

Tech Newsletter Project

Database Administration Guide

Database Location

The project uses a SQLite database.

```
tech_newsletter/  
└─ data/  
    └─ newsletter.db
```

No MySQL or PostgreSQL installation is required.

Database Tables

1. employees

Stores all newsletter recipients.

Column	Type	Description
id	INTEGER	Primary Key
employee_id	TEXT	Employee ID
name	TEXT	Employee Name
email	TEXT	Email Address
department	TEXT	Department Name
status	TEXT	Employee Status (active/inactive)

Current records

```
SELECT COUNT(*) FROM employees;
```

2. sent_articles

Stores every article already emailed. Used to prevent duplicate newsletters.

Column	Type	Description
id	INTEGER	Primary Key
URL	TEXT	Original article URL
title	TEXT	Article title
source	TEXT	News source
category	TEXT	Article category
sent_at	TEXT	Date & Time article was sent

Current records

```
SELECT COUNT(*) FROM sent_articles;
```

3. run_log

Stores every newsletter execution.

Column	Type	Description
id	INTEGER	Primary Key
run_at	TEXT	Pipeline execution time
articles_fetched	INTEGER	Articles collected
articles_sent	INTEGER	Articles emailed
recipients	TEXT	Number/List of recipients
status	TEXT	Execution status
error_msg	TEXT	Error message if failed

Current records

```
SELECT COUNT(*) FROM run_log;
```

Opening the Database

Open DB Browser for SQLite.

Open

data/newsletter.db

The three tables will appear on the left.

View All Employees

```
SELECT * FROM employees;
```

View Active Employees

```
SELECT *  
FROM employees  
WHERE status='active';
```

Search Employee

```
SELECT *  
FROM employees  
WHERE employee_id='EMP001';
```

Search by Email

```
SELECT *
FROM employees
WHERE email='abc@company.com';
```

Add Employee

```
INSERT INTO employees

(employee_id,name,email,department,status)

VALUES

(
'EMP005',
'John Doe',
'john@company.com',
'IT',
'active'
);
```

Update Employee

Change department

```
UPDATE employees

SET department='AI'

WHERE employee_id='EMP005';
```

Disable Employee

Employee will stop receiving newsletters.

```
UPDATE employees

SET status='inactive'

WHERE employee_id='EMP005';
```

Enable again

```
UPDATE employees

SET status='active'

WHERE employee_id='EMP005';
```

Delete Employee

```
DELETE  
  
FROM employees  
  
WHERE employee_id='EMP005';
```

View Sent Articles

```
SELECT *  
  
FROM sent_articles;
```

Latest Sent Articles

```
SELECT *  
  
FROM sent_articles  
  
ORDER BY sent_at DESC  
  
LIMIT 20;
```

Search Sent Article

```
SELECT *  
  
FROM sent_articles  
  
WHERE url='ARTICLE_URL';
```

Delete One Sent Article

```
DELETE  
  
FROM sent_articles  
  
WHERE id=10;
```

Clear Sent Article History

Use only if you want the system to allow previously sent articles again.

```
DELETE FROM sent_articles;
```

View Pipeline History

```
SELECT *  
  
FROM run_log  
  
ORDER BY run_at DESC;
```

View Failed Runs

```
SELECT *  
  
FROM run_log  
  
WHERE status!='ok';
```

Latest Execution

```
SELECT *  
  
FROM run_log  
  
ORDER BY run_at DESC  
  
LIMIT 1;
```

Total Pipeline Executions

```
SELECT COUNT(*)  
  
FROM run_log;
```

Database Integrity Check

```
PRAGMA integrity_check;
```

Expected output

ok

Backup Database

Before making changes, copy

data/newsletter.db

to

```
backup/newsletter.db
```

or use

```
copy data\newsletter.db backup\newsletter.db
```

Restore Database

1. Stop the application.
2. Replace

```
data/newsletter.db
```

with the backup copy.

3. Restart the application.
-

Optimize Database

```
VACUUM;
```

Check Database Size

```
dir data
```

Administrator Checklist

- Verify `newsletter.db` exists.
- Verify all three tables are present.
- Confirm employee records are correct.
- Check `sent_articles` to ensure duplicate prevention is working.
- Review `run_log` after each scheduled execution.
- Back up the database before any manual changes.
- Run `PRAGMA integrity_check;` periodically.